New Attempt

- Due Dec 4, 2024 by 2:59am
- Points 45
- Submitting a text entry box, a website url, or a file upload

Robot Project

Robot Challenge 4

- **Overview:**

  This is the last robot challenge for your team! Begin by meeting with your team to discuss the successes and challenges faced in Challenge 3 and how you plan on wrapping up your project for Challenge 4. Together, program your robot to navigate an unknown maze using a sensor to avoid obstacles. Be sure to record each step of the process and document all relevant details in your digital design notebook. Note that a big part of this challenge is for your robot to function on the day of the demo! If you robot does not function on the day of the demo, you will receive partial credit for submitting a video.

  Reminders:

  1- This is your last chance to incorporate the best design of your team's laser cut design into your robot. Part of your project grade is for just doing that.

  2-Your team is expected to be present on demo day with your robot fully operational, as a portion of the Challenge 4 grade will depend on its functionality during the demo. Recorded videos will no longer be accepted as a substitute for demonstrating functionality on the final demo day.

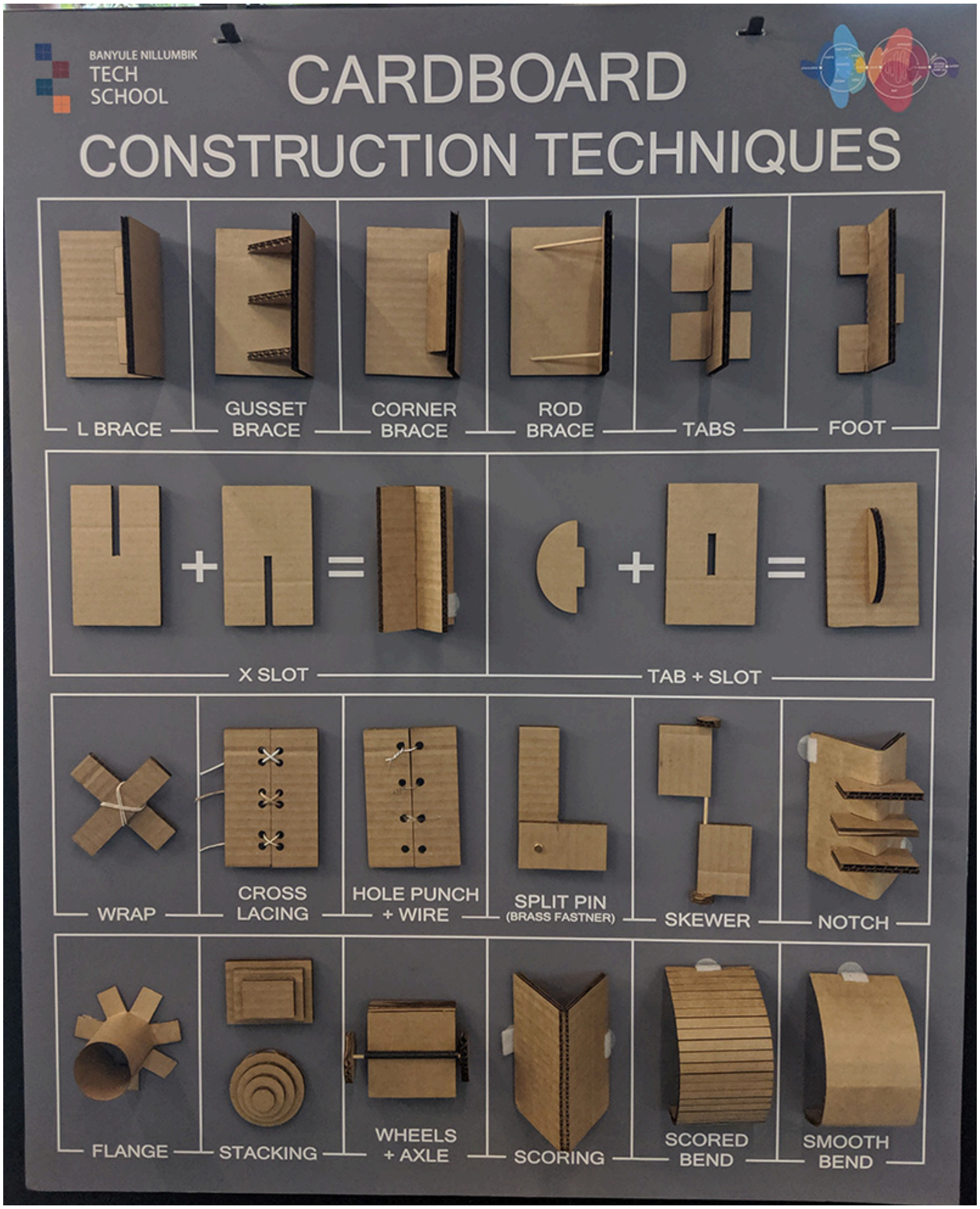
  3- Your Modular Maze should be built and ready for another team to use it.

  4- You should begin working on your individual portfolio using the design history you have been documenting in your digital design notebook.

- **Tasks:**
  - **Task 1: Reflection: As a team, reflect on the followings and record your insights in your design notebook:**
    - You received peer feedback in the form of comments in your notebook and/or on Canvas. You can access them via your Robot Challenge 1 submission page. Review and summarize them in a new "Peer Feedback" section in your digital design notebook. Reflect on the key points mentioned and address the issues identified.
    - Provide a brief explanation of how your team functions and splits the work.

        - Provide a detailed reflection on each team member's contributions to Challenges 3 and 4. If the team member's contributions were not equal, include a clear and specific explanation of the reasons behind the discrepancy.
    - **Task 2**: **Program the robot to drive autonomously in an unknown maze made of carboard, avoiding walls using sensors. You should NOT pre-program your robot.**
        - Take some time to think about how you will change or update your code to handle Robot Challenge 4-- an unknown maze. Draw a flowchart (or write) pseudocode for the autonomous unknown maze challenge. Focus on what might be different between the known maze and unknown maze, and how you might account for some trickery on your classmates' parts.
        - **Include in your digital design notebook:** A paragraph explaining how you coded your approach to the unknown maze
            - You can additionally include a final flowchart for your code using draw.io
            - You might want to consider answers to the following questions:
                - How does your robot sense and react to obstacles? How close does your robot get to an obstacle? What does your robot do when it senses an obstacle?
                - How does your robot adapt to unknown maze conditions? How does your code handle errors or unexpected situations?
                - How did you iterate on your code based on testing results?
                - Did you write any of your own functions? What do they do?
        - Your team is expected to be present on demo day with your robot fully operational. During the demo day (Tuesday, December 3rd) your robot will be assessed in three categories listed below. You will **Submit three videos** of your best attempts from the demo day for each of the categories. <u>Videos recorded anytime other than the demo day will no longer be accepted as a substitute.</u>
            1. Performance in your own maze, on a scale of 1-5
            2. Performance in a random maze of my choosing, on a scale of 1-5
            3. Performance in any maze of your choosing, on a scale of 1-5
            4. *You can include a link to the videos on YouTube, Google drive, or other cloud service if you can't/don't want to upload the actual video*
            5. <u>You will have unlimited attempts to navigate/record each maze during the demo</u>. Each maze will be labeled with the corresponding team name. Please start your video by clearly showing the maze label.
        - Your robot meets the following requirements:
            - Use button/switch/some other trigger to <u>begin the drive program.</u> The reset button does not count as a 'trigger' that begins the program. It resets the entire board.
            - Your robot operated **autonomously** using sensors included in the Sparkfun kit.
            - If you choose to use the distance sensor, note that you have a 3D printed mount to secure the sensor on your robot. You'd need a F to M jumper wire to connect the sensor to your boards.

        - Able to drive
        - Able to turn
        - Able to travel within the boundary without hitting any walls. (Min 10")
        - Navigate the maze from the start to the end.
    - **Task 3: Your completed maze, ready for another team to use with their robot**
        - Build your Modular Maze to scale based on your AutoCAD drawing submitted in Challenge 3 using cardboard and/or any other material in the maker classroom/maker space.
        - If you need to adjust your design, that is OK just remember to create an "as-built" drawing in AutoCAD and include in your digital design notebook. (As-built drawings are a set of drawings that show how something was actually built, as opposed to how it was originally designed)
        - **Include in your design notebook:** Evidence of the building process and a photo of your final maze. Also add an "as-built" AutoCAD drawing if your built maze does not match your drawing from Challenge 3. See image below for some ideas to consider to support the walls of your maze and create interesting features with cardboard!

- 

- **Reminder:** Requirements for the Modular Maze
    - At least three turns (and no more than 5)
    - Walls at least 6 inches tall
    - Able to be sensed by the sensors included in the Sparkfun kit
    - Can be reconfigured quickly (less than 2 minutes)
    - Constraint: Narrowest point at least 10 inches wide (*match the width of the maze from Challenge 3)*

- **Task 4: Record everything in your digital design notebook and share the link on Canvas. Please do NOT email it to Prof. Keyvani.**
    - Make sure the link is accessible and does not require permission. You can test this by copy and pasting the link to an incognito window.
    - Your reflections from task 1
    - A paragraph explaining how you coded your approach to the unknown maze

        - You can additionally include a final flowchart for your code using draw.io
        - You might want to consider answers to the following questions:
            - How does your robot sense and react to obstacles? How close does your robot get to an obstacle? What does your robot do when it senses an obstacle?
            - How does your robot adapt to unknown maze conditions? How does your code handle errors or unexpected situations?
            - How did you iterate on your code based on testing results?
            - Did you write any of your own functions? What do they do?
    - Submit two video of your best attempts (can be from during class or out of class)
        - You can include a link to the video on YouTube, Google drive, or other cloud service if you can't/don't want to upload the actual video
    - Plenty of pictures
    - Meeting minutes
    - Any success or failure
    - Be sure to update your digital design notebook every time you work on your robot project individually or as a team.
    - Modular maze construction evidence
    - A screenshot of your "as-built" dimensioned AutoCAD drawing if your built maze does not match your drawing from Challenge 3.
- **Task 5: Here is everything you need to do to officially complete Cornerstone 1, all in one place:**
    - **After Challenge 4 Demo. take apart your robot (Due 12.4)**
    - **Clean out your team's shelf in the cabinets (Due 12.4)**
    - **Put your Sparkfun kit back together** and keep it in a safe place for next semester
        - If you're going to London, that means taking it with you across the pond!
        - If you find yourself with extra Sparkfun components, bring them to the classroom so that they can be distributed back where they belong
        - And if you suspect you might have switched your kit with someone else's, let me know!
    - **Return things you borrowed (Due 12.4)** (or "borrowed") from the classroom
        - Things like measuring tapes, rulers, calipers, USB-C converters...
    - **Complete late assignments** by 12/4 (Midnight!)
    - **Complete portfolio** by 12/8 (Midnight!)
    - **Complete TRACE** and tell me what worked and/or did not work!

File Submission

As a group, submit the followings:

- Provide an accessible link to your digital design notebook.
- Arduino code(s)

Below you can find the final Demo Day Instruction for Calculating your maze's difficulty score, your robot's performance score and the curved performance scores.

Section 1: Criteria for Determining Maze Difficulty (Score: 0-8)

1. Number of Turns (0-2 points)
   - 3 turns: +1 point
   - 4 or 5 turns: +2 points
     Rationale: More turns require greater navigation effort.
2. Inclusion of Dead Ends (0-2 points)
   - 1 dead end: +1 point
   - Multiple dead ends: +2 points
     Rationale: Dead ends test the robot's ability to detect and stop.
3. Turn Angles (0-1 points)
   - All right-angle turns: 0 point
   - Mix of right and non-right angles (e.g., 45°, obtuse): +1 points
     Rationale: Non-right-angle turns add navigation complexity.
4. Turn Direction (0-2 points)
   - All right or left turns: 0 point
   - Mix of right and left turns: +2 points
     Rationale: mix of left and right turns requires greater decision-making effort.
5. Width Variability (0-1 points)
   - Uniform width: +0 points
   - Includes narrower/wider sections (all ≥10 inches): +1 point
     Rationale: Changing widths require the robot to adapt its movement.

Section 2: Performance Score Rubric

Criteria	Score

The robot successfully completes the maze autonomously: - The robot is turned ON using a proper trigger mechanism, - Navigates all turns without errors (e.g., no collisions or delays). - Stops at the designated endpoint or handles the dead-end appropriately (if applicable).	5
The robot completes the maze autonomously with minor issues: - There is no proper trigger mechanism (button/switch) - One small error, such as a minor delay or hesitation at a turn. - No more than one collision with walls. - Handles endpoint or dead-end correctly.	4
The robot completes the maze with moderate intervention or errors: - Multiple hesitations or delays at turns (e.g., incorrect paths that are self-corrected). - No more than two minor collisions, or one major collision requiring manual intervention. - Requires minimal external correction (e.g., repositioning) but ultimately finishes.	3
The robot fails to complete the maze but demonstrates partial functionality: - Attempts navigation but repeatedly chooses incorrect paths. - Requires frequent external intervention (e.g., repositioning). - Collisions occur often, but the robot displays evidence of programmed behavior (e.g., retries or corrections)	2
The robot shows limited autonomous functionality: - Fails to navigate more than half of the maze independently. - Requires significant external intervention for navigation. - Does not respond to all programmed commands (e.g., ignores turns, fails to detect walls).	1
The robot fails to function meaningfully: - Does not attempt to navigate the maze. - Requires complete external control. - Displays no evidence of programmed functionality.	0

Section 3: Equation for Adjusting Individual Performance Scores

$$Adjusted\ Score = Performance\ Score \times (1 + \frac{Difficulty\ Score}{8} \times 0.5)$$

- Performance Score: The robot's performance score based on the rubric for one maze (out of 5)
- Difficulty Score: Determined based on the rubric for the same maze (out of 8)
- Example Calculations for a maze:

Difficulty =4 (out of 8), Performance = 4 (out of 5)

Challenge 4

Criteria	Ratings		Pts
Code upload	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Adjusted Performance Score-own maze Only members present during challenge 4 demo can get credit for this part.	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Adjusted Performance Score-maze of choice Only members present during challenge 4 demo can get credit for this part.	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Adjusted Performance Score-Random maze Chosen by professor or the team Only members present during challenge 4 demo can get credit for this part.	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Constructed Cardboard maze	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Digital Design Notebook	**10 pts** **Full Marks**	**0 pts** **No Marks**	10 pts
Lasercut robot part added to the robot	**10 pts** **Full Marks**	**0 pts** **No Marks**	10 pts
			Total Points: 45